

Pier Felice Balestrucci

Curriculum Vitae

Turin, Italy

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🌐 pierbale.github.io

Work Experience

- Nov 2025-Oct 2026 **Research Fellow**
Department of Computer Science - University of Turin, Turin, Italy
Research project: Understanding, Generation and Management of Pragmatically Constrained Utterances in Dialogue Systems through Perspective-Aware Language Modeling
Scientific Leads: Prof. Alessandro Mazzei, Prof. Valerio Basile
- Oct 2025-Sep 2026 **Contract Lecturer**
Department of Computer Science - University of Turin, Turin, Italy
Course: *Intelligent Systems*, Master's Degree in Strategic and Military Sciences
- Nov 2022-Oct 2025 **Doctoral Researcher**
Department of Computer Science - University of Turin, Turin, Italy
Research on personalized dialogue systems and accessibility technologies
Funded by a national doctoral scholarship (Ministry of University and Research – MIUR)
- Jun–Oct 2022 **Data Scientist**
MA Computer S.r.l., Turin, Italy
Data processing, cleaning, and preparation on a large-scale dataset related to product purchases from a multinational company. Designed a statistical supply-chain optimization module after cleaning and normalizing a large-scale multinational purchasing dataset. Worked in collaboration with a mid-sized data science team.
- Jan–Jun 2022 **Data Analyst (Internship)**
Accenture, Milan, Italy
Contributed to the analysis of financial transactions in a banking context, focusing on anomaly detection and security validation. Applied supervised machine learning algorithms to identify suspicious activity. Worked in collaboration with a mid-sized data analytics team.
- Dec 2020–Aug 2021 **Student Collaborator**
Department of Computer Science - University of Turin, Turin, Italy
Part-time collaboration funded through a university scholarship. Preparation of accessible educational materials for students with disabilities

Education

- Nov 2022 - Oct 2025 **PhD in Computer Science**, *defense expected in Sep 2026*, University of Turin – Turin, Italy
Research topic: Personalization strategies to improve communication in dialogue systems
Supervisors: Prof. Alessandro Mazzei, Prof. Luca Anselma

- Nov 2019 - **Master's Degree in Artificial Intelligence – 110/110 with honors**, Artificial Intelligence and Information Systems, University of Turin – Turin, Italy
Thesis: Voice-based interaction for navigating mathematical expressions
Supervisors: Prof. Alessandro Mazzei, Prof. Luca Anselma
- Sep 2016 - **Bachelor's Degree in Computer Science – 110/110 with honors**, University of Bari – Bari, Italy
Oct 2019
Thesis: "Pepper for Tourism" – Design and development of an information system for tourism in Apulia using the social robot Pepper
Supervisor: Prof. Berardina De Carolis, Prof. Cataldo Musto

Research and Training Experiences

- Sep 2024-Feb 2025 **Research Internship at Institute of Formal and Applied Linguistics, Charles University (Prague, Czechia)**
Supervisor: Prof. Ondřej Dušek
Focus: Personalization of real-world dialogue corpora for different user profiles using Large Language Models (LLMs)
<https://ufal.mff.cuni.cz/home-page>
- Jun 2024 **Lectures on Computational Linguistics 2024, organized by AIxIA and AILC**
Topics: Cognitive Evaluation of LLMs, Lexical Semantic Change, Digital Humanities, Prompting
<https://www.ai-lc.it/en/lectures-2/lectures-2024/>
- Jul 2023 **European Summer School in Logic, Language and Information (ESSLLI)**
Topics: Formal Language Theory, LLMs, Computational Creativity, Semantic Representations
<https://2023.esslli.eu>
- Mar 2023 **Bertinoro International Spring School on AI (BISS)**
Courses: Hybrid Quantum Computing, AI for Health, Neural MT, Programming Language Design
<https://cs.unibo.it/projects/BISS/2023/program/>

Short List of Publications

- [1] Pier Felice **Balestrucci**, Ondrej Dusek, Luca Anselma, Alessandro Mazzei, et al. "Can Large Language Models Personalize Dialogues to Generational Styles?" In: *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, 2025.
- [2] Michael Oliverio, Pier Felice **Balestrucci**, Luca Anselma, Alessandro Mazzei, et al. "AIML+: Enhancing AIML for the Educational Domain Through Frames and Large Language Models". In: *LECTURE NOTES IN COMPUTER SCIENCE* (2025), pp. 176–189. DOI: https://dx.doi.org/10.1007/978-3-031-98459-4_13.
- [3] Michael Oliverio, Pier Felice **Balestrucci**, Alessandro Mazzei, and Valerio Basile. "WebNLG-IT: Construction of an aligned RDF-Italian corpus through Machine Translation techniques". In: *Findings of the Association for Computational Linguistics: ACL 2025*. 2025, pp. 12073–12083.
- [4] Pier Felice **Balestrucci**, Silvia Casola, Soda Lo, Valerio Basile, Alessandro Mazzei, et al. "I'm sure you're a real scholar yourself: Exploring Ironic Content Generation by Large Language Models".

In: *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, 2024, pp. 14480–14494. DOI: <https://dx.doi.org/10.18653/v1/2024.findings-emnlp.847>.

- [5] Pier Felice **Balestrucci**, Elisa Di Nuovo, Manuela Sanguinetti, Luca Anselma, Cristian Bernareggi, Alessandro Mazzei, et al. “An Educational Dialogue System for Visually Impaired People”. In: *IEEE ACCESS* (2024), pp. 1–20. DOI: <https://dx.doi.org/10.1109/access.2024.3479883>.

Research Projects

- 2024–2026 **Multilingual Perspective-Aware NLU** – in partnership with Amazon Alexa
Research project focused on perspective-sensitive natural language understanding across multiple languages. Designed and evaluated LLM-based pipelines for irony generation and detection in multilingual, perspective-aware NLU settings, contributing to research outputs in NLP venues.
- 2021–2024 **NoVAGraphS – Non-Visual Access to Graphical Structures**
<http://www.integr-abile.unito.it/progetto-novagraphs/>
Interdisciplinary research initiative on assistive dialogue systems for blind and visually impaired students, with a focus on non-visual interaction with STEM content. Contributed to the design and implementation of multimodal dialogue interfaces and user modeling strategies.

Public Engagement

- 2023 **FundsTOgether – Crowdfunding for Research and Innovation**
Participation in a university-wide alternative finance program for research and innovation, implemented through crowdfunding. The initiative aims to fund projects with strong social and environmental impact by engaging a wide audience.
- 2023 **MatematlcA senza barriere – Crowdfunding Campaign**
Participation in the “MatematlcA senza barriere” crowdfunding campaign, part of the FundsTOgether program. Activities included training, project selection, and the launch and management of the campaign.
- 2022 **SpeechMate – POV Call, Laboratorio Polin**
Participation in the POV Call with the SpeechMate project, involving collaborations with associations and companies interested in accessibility technologies.
- 2022 **Making Science Inclusive – Workshop, Laboratorio Polin**
Contributed to the organization of a workshop addressing problems, solutions, and proposals for improving the accessibility of scientific texts and publications.

Teaching

- 2023–2024 Tutoring – Programmazione II A, B, C (Cod. inf0330), 40 hours
Bachelor’s Degree in Computer Science, University of Turin
- 2023–2024 Tutoring – Tecnologie Web Approcci Avanzati, 40 hours
Master’s Degree in Communication, ICT and Media, University of Turin
- 2025–2026 Tutoring – Tecnologie Web Approcci Avanzati, 40 hours
Master’s Degree in Communication, ICT and Media, University of Turin

Language Skills

Italian Native speaker

English Upper-intermediate – strong reading, writing, and conversational skills